

# Software for the Analysis of Atomic Spectra (SAAS)

## User Manual

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April 13, 2026

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# 1 Introduction

The Software for the Analysis of Atomic Spectra (SAAS) is a unified suite designed for experimental atomic physics. It replaces a diverse set of legacy scripts (notably `pybranch.py`) with a modern, modular application written in Python 3.

SAAS provides a structured environment for managing spectroscopic data, identifying spectral lines, and calculating fundamental atomic properties such as **Branching Fractions (BFs)** and **Transition Probabilities (Einstein A-coefficients)**.

## 2 Core Architecture

### 2.1 HDF5 Project Management

Unlike legacy workflows that rely on scattered ASCII files, SAAS utilizes a single **HDF5 (.h5)** project file. This format acts as a container for:

- Raw and calibrated spectra (NSO and FITS formats).
- Experimental linelists (.lin and .txt).
- Master energy level tables and theoretical calculations.
- Versioned analysis results and metadata.

### 2.2 The Main Interface

The main window consists of an **HDF5 Tree Browser** on the left and a **Data Preview/Metadata Pane** on the right.

**Tree Selection:** Click any item to see a quick preview (first 200 rows) or a spectral plot.

**Metadata View:** The second tab displays all HDF5 attributes associated with the selected object.

**Full Table Viewer:** Right-click any dataset in the tree and select **Full Table**. This opens a searchable, filterable window containing the entire dataset, essential for navigating large experimental linelists.

## 3 Installation and Build

### 3.1 System Requirements

SAAS is designed for Linux environments (GNOME/Wayland). It requires:

- Python 3.8 or higher.
- Dependencies: `pandas`, `numpy`, `h5py`, `matplotlib`, `PyQt5`, `click`, `astropy`.

### 3.2 Building the Executable

A build script `build.sh` is provided to create a standalone binary:

```
chmod +x build.sh
./build.sh
```

This packages the code and libraries into a single file in the `dist/` directory and generates a Linux `.desktop` file for menu integration.

## 4 Data Import

SAAS provides dedicated wizards to ensure data integrity and schema compliance.

### 4.1 Spectra and Linelists

- **Main Spectrum:** Imports .raw/.dat files with their .hdr headers.
- **Calibration Spectrum:** Associates a standard lamp spectrum with an existing main spectrum to facilitate intensity calibration.
- **Binary Linelists:** Directly parses Xgremlin .lin files, automatically applying damping parameter transformations.

### 4.2 Generic Table Wizard

For Energy Levels and Previous Identifications, the **\*\*Import Wizard\*\*** allows users to:

1. Preview ASCII/CSV files.
2. Assign columns to specific fields (e.g., mapping a column to `lifetime_unc_frac`).
3. Store source metadata (author, date, notes) directly in the project file.

## 5 Interactive Branching Fraction Analysis

The analysis module (accessible via **Ctrl+R**) is the heart of SAAS.

### 5.1 The Analysis Workflow

1. **Level Selection:** Choose a master levels file and select an upper level of interest. SAAS filters the Previous Identifications list to show transitions originating from that level.
2. **Data Selection:** Select the experimental linelists for comparison. *Note: Checking a linelist automatically enables its corresponding raw spectrum for plotting.*
3. **Normalization:** Select a strong line observed in most spectra. Right-click and select **Normalize to Reference** to set its intensity to 1000 and rescale all other lines.
4. **Transfer Calibration:** If a spectrum lacks the primary reference line, right-click a common “Transfer Line” to bridge the intensity scales between spectra.

### 5.2 Outlier Highlighting

SAAS performs real-time outlier detection. If an individual intensity measurement deviates by more than **3-sigma** from the weighted mean, the cell is highlighted in **\*\*light red\*\***. Sigma is calculated as the root-sum-of-squares of the absolute individual uncertainty and the mean uncertainty.

## 6 Physical Calculations

### 6.1 Uncertainty Model

SAAS implements the model established by Sikström et al. (2002). The total absolute uncertainty of an individual measurement is derived from:

1. **Statistical Uncertainty** ( $U_{stat}$ ):

$$U_{stat} = \frac{1.5}{SNR \cdot \sqrt{n}} \quad (1)$$

where  $n = \text{FWHM}/\text{resolution}$ .

2. **Calibration Uncertainty** ( $Unc_{cal}$ ): This term accounts for the increasing error as one moves away from the spectrum's strongest calibrated line ( $\sigma_{maxI}$ ):

$$Unc_{cal} = \text{CalUncPer1000} \cdot \frac{|\sigma - \sigma_{maxI}|}{1000} \quad (2)$$

## 6.2 Weighted Mean and Constraints

Lines are weighted by  $1/Unc_{total}^2$ . To prevent strong lines from disproportionately biasing the results, uncertainties are clipped to a minimum of 6% (a weight cap of 555).

## 6.3 Residual Estimation

Unobserved decay channels are accounted for by a “residual” fraction calculated from theoretical transition probabilities ( $A_{theo}$ ):

$$\text{Fraction}_{\text{resid}} = \frac{\sum A_{\text{unobserved}} \cdot \tau_{\text{lifetime}}}{1000} \quad (3)$$

A default uncertainty of 50% is assigned to the residual.

# 7 Reporting and Export

## 7.1 The Rule of 20

To maintain scientific rigor, results are rounded according to the metrological **\*\*Rule of 20\*\***:

- If the first significant digit of the absolute uncertainty is **1**, the value is rounded to two significant digits of the uncertainty.
- If the first significant digit is **2 or greater**, the value is rounded to one significant digit of the uncertainty.

## 7.2 Export Formats

**LaTeX**: Generates a `tabular` environment with all percentage signs and special characters properly escaped for direct insertion into papers.

**MRT (Machine Readable Table)**: Exports using the `astropy.io.ascii.mrt` library. This format is the standard for AAS journals (ApJ, AJ). SAAS provides both **\*\*Truncated\*\*** (matching the UI) and **\*\*Raw\*\*** (full precision) MRT export options.

# 8 Troubleshooting

- **Visual Artifacts**: If random colored patterns appear on Linux, SAAS automatically forces the `xcb` (X11) backend and software rendering (`LIBGL_ALWAYS_SOFTWARE=1`) to ensure stability on Wayland.
- **File Access**: If HDF5 files on network drives show locking errors, run:  
`export HDF5_USE_FILE_LOCKING=FALSE` in your shell.